

Nandan Bharatkumar Parikh

(404)-883-8113 | nparikh44@gatech.edu | [linkedin.com/in/nandanbparikh](https://www.linkedin.com/in/nandanbparikh) | github.com/lelouch0204

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Master of Science in Computer Science | GPA: 4.0/4.0

August 2025 – May 2027

- Relevant Coursework: Computer Vision, Machine Learning, Efficient ML
- Graduate Teaching Assistant - Compilers and Interpreters (Spring 2026): Grading assignments and leading weekly office hours to clarify concepts and debug student implementations.
- Graduate Student Researcher - Developing diffusion-based image-to-video generation models (Hays Lab).

Birla Institute of Technology and Science, Pilani

Pilani, India

Bachelor of Engineering in Computer Science

August 2019 – May 2023

- Relevant Courses: Operating Systems, Computer Networks, DBMS, Compiler Construction, OOP

EXPERIENCE

Incoming AI/ML Intern

Summer 2026

Apple | Siri Core Modeling

Cupertino, CA

- Will work on agentic AI, tool calling, and planning for large language models

Software Engineer II (SDE-2)

July 2023 – July 2025

Flipkart (Walmart) | Hotels Search

Bengaluru, India

- **Hotel Open Search:** Built LLM-powered natural language search achieving 92% recall with sub-100ms p95 latency, translating free-form queries into structured retrieval via LangChain4j and Elasticsearch
- **Vector Search Infrastructure:** Evaluated vector databases (Qdrant, Pinecone) and optimized existing Elasticsearch cluster to avoid \$20K annual infra costs while meeting search performance requirements
- **Scaling Systems:** Designed and implemented scalable backend services in Java/Spring Boot, improving throughput by 5× (5k → 24k requests/sec) via performance tuning, load testing, threadpool & JVM optimization.
- **Search Architecture Revamp:** Redesigned search APIs and caching layers to reduce p95 latency by 50% (2s → 900ms), cutting database load by 70% through Redis-backed multi-tier caching.
- Improved reliability of offline indexing pipelines by adding execution-level observability, preventing silent failures and downstream search inconsistencies.
- Reduced zero search results by 35% by building a click-tracking pipeline using BigQuery for autosuggest optimization and enabling support for informally-defined geographic areas.
- Promoted early from SDE-I for exceptional backend architecture and scalability contributions

Research Intern | [Publication]

November 2022 – June 2023

Harvard University

Boston, MA

- Implemented an end-to-end microscopy image deblurring pipeline with OpenCV, PyTorch, and CUDA.
- Achieved 20× faster inference over SoTA methods through a graph-based lightweight attention module while maintaining reconstruction quality (PSNR 33.23, SSIM 0.88).
- Work accepted for poster presentation at Microscopy and Microanalysis (peer-reviewed).

PROJECTS

Distributed Training Implementation (FSDP & DDP) | PyTorch, NVIDIA H100s, CUDA

GitHub

- Implemented **Fully Sharded Data Parallel (FSDP)** and Pipeline Parallelism on **NVIDIA H100** clusters to scale transformer training.
- Achieved **96.8% scaling efficiency** (near-linear) across multi-GPU nodes by optimizing pipeline micro-batching and masking communication overhead.
- Benchmarked FSDP vs. DDP, maintaining **99.8% baseline throughput** while reducing memory fragmentation.

Efficient LLM Inference & Quantization | PyTorch, Llama-3, SmoothQuant

GitHub

- Enabled **W8A8 inference** on Llama-3 via SmoothQuant, preserving FP16 fidelity (6.24 PPL) despite 40x activation outliers
- Analyzed Attention Sinks, finding initial tokens capture 95% probability mass in deep layers to validate StreamingLLM mechanics
- Profiled KV Cache dynamics, quantifying linear memory growth at 0.125 MB/token for long-context optimization

Political Bias Detection | RoBERTa, SimCSE, SetFit

Website

- Fine-tuned **RoBERTa/SetFit** on **300k+ corpus**, boosting baseline F1 by **9%** via few-shot learning.
- Implemented **SimCSE (Contrastive Learning)** to optimize embedding alignment and ideological clustering.
- Benchmarked **6 algorithms** on **GDELT data** to quantify and visualize bias spread across media sources.

TECHNICAL SKILLS

Languages: Java, Python, C++, C, SQL, JavaScript

Backend & Architecture: Spring Boot, Microservices, Event-Driven Architecture, Multithreading

Data & Messaging: Kafka, Redis, Aerospike, Elasticsearch, MongoDB, SQL

Cloud & DevOps: GCP, Docker, Kubernetes, CI/CD